PG – 376

III Semester M.C.A. (Two Years Course) Examination, May 2024
(CBCS) (2021-22)
COMPUTER SCIENCE
MCA 3E3/MSC 3E3 : Cryptography and Network Security (Elective)

Time : 3 Hours Max. Marks : 70

***Instruction** : Answer **any five** from Section – **A** and **any four** from Section – **B**.*

SECTION – A

Answer **any five** of the following. **Each** question carries **six** marks. **(5×6=30)**

1. For a plaintext CRYPTOGRAPHY, obtain the ciphertext using transposition cipher and the following key pattern

   $1 \rightarrow 4, 2 \rightarrow 8, 3 \rightarrow 1, 4 \rightarrow 5, 5 \rightarrow 7, 6 \rightarrow 2, 7 \rightarrow 6$ and $8 \rightarrow 3$.

2. Describe the structure of Simplified DES (S-DES) and its key components. How does S-DES differ from the full Data Encryption Standard (DES) ?

3. Define block cipher. Explain CBC mode of encryption and decryption with a diagram.

4. Describe common cryptographic attacks such as brute force attacks, ciphertext-only attacks, and known-plaintext attacks. Provide examples of countermeasures that can be implemented to mitigate these threats.

5. Explain Chinese Remainder theorem and find X for the given set of congruent equation using CRT

   $X \equiv 2 \pmod 3$ $X \equiv 3 \pmod 5$ $X \equiv 2 \pmod 7$.

6. Discuss the Diffie-Hellman key exchange algorithm with its merits and demerits.

7. Explain Intrusion Detection System (IDS) in detail with suitable diagrams.

8. Explain SSL architecture with a suitable SSL protocol stack.

P.T.O.

SECTION – B

Answer **any four** of the following. **Each** question carries **10** marks. **(4×10=40)**

9. Summarize OSI security architecture model with neat diagram. Discuss the significance of random numbers in network security.

10. Explain attacks of RSA cryptosystem. Perform encryption and decryption using RSA algorithm for the following p = 7, q = 11; e = 17; m = 8.

11. Explain the concept of hash functions in cryptography. What are the essential properties of a cryptographic hash function, and how are hash functions used for data integrity, digital signatures, and password storage ?

12. Describe the process of digital signatures and their importance in ensuring message authenticity and non-repudiation. How do digital signatures differ from electronic signatures ?

13. What is Kerberos ? Explain how it provides authenticated services. Explain the format of the X.509 certificate.

14. Define Firewall. List out the types of firewalls. Explain packet filtering firewall with suitable filtering rules.

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